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DEPARTMENT OF TEACHER EDUCATION
SCHOOL OF EDUCATION AND LEADERSHIP
COLLEGES OF EDUCATION
END OF SEMESTER ONE EXAMINATIONS FOR LEVEL 300, 2023/2024
B.ED. PROGRAMME

COURSE CODE: TEJS 333
COURSE TITLE: WEATHER AND CLIMATE

Instruction: Answer **all** questions in Section A and any **three** questions in Section B.

Time: 2 hours

SECTION A

[25 Marks. Each Question is 1 Mark]
Answer all the questions in this section.

1. The following climatic zones were derived by Aristotle, EXCEPT _____
   A. continental
   B. frigid
   C. temperate
   D. torrid
2. Which of the following lower portions orlayers do aircrafts fly to avoid turbulent conditions and increase in atmospheric drag?
   A. mesosphere
   B. stratosphere
   C. thermosphere
   D. troposphere
3. Which of the following instruments is used to measure atmospheric pressure?
   A. Betrometer
   B. Seismograph
   C. Barometer
   D. Wind vane
4. All the following are weather conditions, except _____
   A. cloudiness
   B. rainy
   C. shiny
   D. sunny

5. The earth's atmosphere contains the following, except…
   A. dust
   B. mud
   C. sea salt
   D. volcanic ash
6. As altitude increases in the atmosphere, air pressure
   A. decreases
   B. increases
   C. stabilizes
   D. stagnates
7. Which of these clouds contribute to thunderstorms?
   A. Cirrusnimbus
   B. *Cumulonimbus*
   C. Cumulus
   D. Nimbus
8. Which of these is the dominant permanent atmospheric gas?
   A. carbon dioxide
   B. hydrogen
   C. oxygen
   D. water vapour
9. When clouds that form at ground level and raise ambient humidity to its maximum to decrease visibility is termed as ….
   A. Fog
   B. Hail
   C. shower
   D. Snow
10 The heating of the atmosphere from the earth's surface occurs mainly through a process known as
   A. Shortwave radiation
   B. Surface albedo
   C. Convection
   D. Diffused radiation
11. A standard weather station would include
   i. Anemometer
   ii. Stevenson screen
   iii. Telescope
   iv. Wind vane
   A. i, ii, iii
   B. i, ii, iv
   C. i, ii, iv
   D. i, iii, iv

12. During the harmattan, there is usually a large difference in the readings of the wet and dry bulbs thermometers because _____
    A. Air is saturated
    B. Relative humidity is high
    C. Temperature is high
    D. Relative humidity is low
13. In Koppen's classification of climates, AW refers to _____
    A. Tropical
    B. Steppe
    C. Mediterranean
    D. Humid sub-tropical
14. The difference between in incoming and outgoing shortwave and longwave radiation is represented in radiation balance equation as _____
    A. Rnet= SWnet + LWnet
    B. Rnet= SWnet - LWnet
    C. Rnet= LWnet - SWnet
    D. SWnet= Rnet + LWnet
15. The balance between incoming energy from the sun and outgoing energy from earth constitute _____
    A. Solar radiation
    B. Energy balance
    C. Terrestrial radiation
    D. Energy saving
16. When Ghana's atmosphere becomes dry, hazy, and dusty with reduced visibility accompanied by high day and low night temperatures particularly between November and February, it means Ghana is under the influence of _____
    A. Tropical continental air mass
    B. Tropical maritime air mass
    C. Equatorial maritime air mass
    D. North East Trade Wind
17. High temperatures and heavy rainfall distributed throughout the year are characteristics of which climatic region?
    A. Tropical monsoon climate
    B. Equatorial climate
    C. Mediterranean climate
    D. Tundra climate
18. Which of the following does not explain why hot deserts have high temperatures?
    A. Presence of clear cloudless sky
    B. Presence of dry air
    C. Intense insolation
    D. Ascending air
19. Which of the following is a direct effect of atmospheric erosion?
    A. Fall in sea level
    B. Atmospheric instability
    C. Formation of acid rain
    D. Growth of thick vegetation

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20. They are composed of water droplets; forming uniform sheet or grey or watery-looking clouds, partly or totally covering the sky. They normally appear when it is about to rain. This description is suitable for _____
    A. Altocumulus
    B. Altostratus
    C. Cirrostratus
    D. Nimbostratus
21. Greek classification of climate is based completely on _____
    A. Rainfall
    B. Soil
    C. Temperature
    D. Vegetation.
22. Winds are caused by differences in_____
    A. Air pressure
    B. The thermosphere
    C. Oxygen
    D. Gravity
23. The following are all sources of carbodioxide into the atmosphere except _____
    A. Respiration
    B. Transpiration
    C. Fossil fuel
    D. Natural gas
24. The energy that is trapped in parked car in the sun during the day is known as _____
    A. Short waves
    B. Infrared radiation
    C. Long waves
    D. Visible light
25. Which of the following is not a form of precipitation?
    A. Rainfall
    B. Snow
    C. Hail
    D. Haze

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SECTION B

[75 Marks]

Answer any three questions in this section

1 a. In a tabular form give four differences between weather and climate [9 marks]

b. Name the unit of measurement for each of the following climatic elements.

i. Radiation [2marks]
ii. Rainfall [2marks]
iii. Atmospheric Pressure [2marks]
iv. Humidity [2marks]
v. Wind speed [2marks]

c. Explain three (3) factors affecting weather and climate [6marks]

2. a. Explain the term climate **[4 marks]**

b. Discuss **four** human activities that can lead to climate change. **[12 marks]**

c. Examine **three** effects of climate change. **[9 marks]**

3. a. Explain the term atmosphere **[4 marks]**

b. Examine three characteristics of the atmosphere. **[9 marks]**

c. Explain four importance of the earth's atmosphere. **[12 marks]**

4. a. What is evaporation? **[4 marks]**

b. Outline **three** factors affecting the rate of evaporation in an area. **[9 marks]**

c. Explain each of the following terms.

i. relative humidity
ii. specific humidity
iii. absolute humidity **[12 marks]**

5. Define air masses and explain their characteristics and sources of the types of air masses

[25 marks]